Venkata Lakshmi Parimala Pasupuleti

Software Development Engineer · Distributed Systems · Cloud Services · Observability · Java · Python · AWS · Microservices · Event-Driven · Full SDLC · Agile

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PROFESSIONAL SUMMARY

Software Development Engineer with 8+ years building production-grade distributed systems, cloud services, and data pipelines at scale. At Entrata, owned full-lifecycle delivery of backend systems serving 100K+ monthly active users: event-driven architecture (RabbitMQ), multi-tenant PostgreSQL, AWS cloud services, microservices redesign, and observability instrumentation across 120+ Agile sprints. Built AgentProbe: open-source AI evaluation infrastructure with trace-level observability, CI-integrated regression harness, and distributed model routing. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models, data pipelines, and APIs surfacing insights from raw business data into application-facing reporting layers.

George Mason University Aug 2024 – Aug 2025

Research Associate + Teaching Assistant

- Built ML prediction pipelines (Python, R) across 190+ countries (R^2=0.977); deployed inference API (PyTorch + HuggingFace); Isolation Forest anomaly detection across 3 research pipelines. Delivered technical workshops to 600+ students across Python programming and database design.

Entrata (Enterprise SaaS / Real Estate Technology) Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Owned and delivered features across the full software lifecycle for 7 years** — designed, implemented, tested, deployed, and operated backend systems and cloud services for 100K+ monthly active users across 120+ Agile sprints. Worked backwards from customer and compliance requirements to build solutions built to last.
- **Built fault-tolerant event-driven cloud services (RabbitMQ)** processing thousands of account state updates — retry logic, idempotency guarantees, designed for correctness under concurrent events; owned through production incidents and drove issues to permanent root-cause fixes. 99.5% uptime across the ownership period.
- **Designed and maintained distributed multi-tenant systems** on AWS (S3, EC2, RDS) — schema design supporting data isolation across 100K+ units; optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, and partitioning; microservices redesign (Docker) yielding 50% scalability improvement and 40% deployment time reduction.
- **Instrumented production services with observability tooling** (OpenTelemetry, Prometheus, Grafana) — monitoring, alerting, and operational dashboards for mission-critical data pipelines; reduced debugging time by 40%; maintained bug count below 10 for 6+ consecutive months.
- Mentored 5+ engineers; conducted 300+ peer code reviews; raised engineering bar through standards, debugging tools, and best practices across a globally distributed team.

Entrata India, Pune Jul 2017 – Sep 2018

Associate Software Engineer

- Built backend modules (Java, Python, PostgreSQL); led AWS S3 cloud migration; redesigned document storage architecture; managed data pipelines across hundreds of thousands of records.

PROJECTS

AgentProbe: Distributed AI Evaluation + Observability Infrastructure OBSERVABILITY CI-INTEGRATED

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · OpenTelemetry | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend, trace-level observability, 5-metric evaluation engine, CI-integrated regression harness with golden datasets, distributed model routing across OpenAI/Anthropic/Groq/Ollama. Built to measure reliability and accuracy of AI-powered services before they reach production users.

Leasel: 8-Agent AI Platform - Distributed Workflow + HITL LANGGRAPH FAULT-TOLERANT

Python · LangGraph · FastAPI · SQLite · OpenAI API | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Distributed 8-agent workflow with durable state management, conditional routing, fault-tolerant HITL escalation, and citation-grounded RAG over document corpus. 93.6% precision, 87.9% F1 across 1,240 documents. Designed for correctness: LLM reasons, deterministic logic makes every regulated decision.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur 2017

SKILLS

Languages: Java (Spring Boot), Python (8+ yrs), JavaScript, TypeScript, SQL, PHP, Bash, R — FastAPI, Django, React.js, REST APIs

Distributed Systems & Backend: Microservices, Event-Driven Architecture (RabbitMQ, Apache Kafka), Distributed Systems, Fault-Tolerant Design, Idempotency, Multi-Tenant Architecture, Data Isolation, PostgreSQL, Redis, Celery, Temporal, Docker, Kubernetes

Cloud & Observability: AWS (S3, EC2, RDS, Glue, Lambda, Step Functions, Bedrock), OpenTelemetry, Prometheus, Grafana, CI/CD (GitHub Actions), Terraform, Pulumi, Performance Engineering, Scalability

AI & Data: LangGraph, OpenAI API, Anthropic API, RAG Pipelines, LLM Evaluation, MCP, ChromaDB, PyTorch, HuggingFace, scikit-learn, NumPy, Pandas, Tableau